

E-Commerce Sales Performance Analysis (SQL + Power BI)

Project Overview

This project presents a complete end-to-end business intelligence solution for an e-commerce platform using PostgreSQL and Power BI. The objective of the project is to analyze sales performance, customer behavior, seller efficiency, delivery operations, and retention trends to uncover actionable business insights.

The project combines:

- SQL for data modeling, transformation, and business analysis
- Power BI for dashboarding and visualization
- DAX for KPI calculations and cohort analysis

The final dashboard was designed using an Amazon-inspired theme with interactive slicers, KPI cards, trend analysis, and cohort retention heatmaps.

Business Objective

The main goal of this project is to help e-commerce businesses understand:

- Which customers generate the highest revenue
- Which categories and sellers perform best
- How discounts affect sales
- How delivery performance impacts operations
- How customer retention changes over time

The dashboard transforms raw transactional data into meaningful business insights that can support strategic decision-making.

Dataset Overview

The dataset contains transactional e-commerce records including:

- Orders
- Products
- Sellers
- Customer purchases
- Payment methods

- Delivery information
- Discounts
- Device usage
- Customer locations

Due to the large dataset size, the dataset files are not included in this repository.

Dataset Source: <https://www.kaggle.com/datasets/sharmajicoder/amazon-e-commerce?resource=download>

Tools & Technologies Used

Tool	Purpose
PostgreSQL	Data modeling and business analysis
Power BI Desktop	Dashboard creation and visualization
DAX	KPI calculations and cohort analysis
Power Query	Data transformation
GitHub	Version control and project hosting

Data Modeling

The project follows a star-schema inspired data model.

Fact Table

orders

Stores transactional order-level information including:

- order_id
- user_id
- product_id
- seller_id
- final_price
- purchase_date
- payment_method
- delivery_status

Dimension Tables

products

Contains product-level information:

- category
- subcategory
- brand
- rating

sellers

Contains seller-level information:

- seller_id
- seller_rating

Relationships

- orders.product_id → products.product_id
- orders.seller_id → sellers.seller_id

This structure improves query efficiency and dashboard performance.

SQL Analysis Performed

The SQL layer of the project includes multiple business analysis queries.

1. Top 20% Customer Revenue Contribution

Used window functions and ranking analysis to determine how much revenue is generated by the top 20% customers.

Insight

A small percentage of customers contribute the majority of revenue.

2. Repeat vs One-Time Customer Analysis

Customers were categorized into:

- One-time customers

- Repeat customers

Insight

Repeat customers generated a significant portion of overall revenue.

3. Monthly Revenue Trend Analysis

Revenue was aggregated monthly to identify:

- growth patterns
 - seasonal trends
 - revenue fluctuations
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4. Category Contribution Analysis

Analyzed product categories to identify:

- highest revenue-generating categories
- category-wise contribution percentages

Insight

Electronics category generated the highest revenue.

5. Top Seller Performance Analysis

Measured seller performance using:

- total orders
- revenue generated

Insight

A few sellers contributed significantly higher revenue than others.

6. Delivery Performance Analysis

Calculated average delivery times across different delivery statuses.

Insight

Most deliveries were completed within 3–4 days.

7. Discount Impact Analysis

Grouped discounts into:

- No Discount
- Low Discount
- High Discount

Insight

Medium-level discount strategies generated the highest revenue.

8. Customer Cohort Retention Analysis

Performed cohort analysis to measure customer retention over time.

Insight

Customer retention dropped significantly after Month 2.

Power BI Dashboard Features

The Power BI dashboard includes:

KPI Cards

- Total Revenue
- Average Order Value
- Top 20% Revenue Contribution
- Average Shipping Days

Visualizations

- Monthly Revenue Trend
- Category Contribution Analysis
- Repeat vs One-Time Customers
- Delivery Performance Analysis
- Discount Impact Analysis

- Top Sellers Analysis
- Cohort Retention Heatmap

Interactive Features

- Slicers and filters
 - Dynamic visual interactions
 - Responsive dashboard layout
 - Amazon-inspired custom UI theme
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Dashboard Design

The dashboard uses a modern Amazon-inspired theme with:

- Dark navy sidebar
- Gold accent colors
- White background layout
- Rounded cards and visuals
- Interactive slicers with custom icons

Color Palette:

• **232F3E**

• **CEA968**

• **E4C083**

• **08AAE3**

- White
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Key Business Insights

- Top 20% customers contribute nearly 68% of total revenue.
- Repeat customers significantly outperform one-time customers.
- Electronics category generated the highest sales contribution.
- Medium discount strategies drove maximum revenue.
- Customer retention decreases sharply after the second month.
- Delivery performance remained relatively efficient across orders.

Challenges Faced

During the project, several challenges were addressed:

- Building proper cohort retention calculations in Power BI
- Structuring fact and dimension tables correctly
- Managing Power BI date hierarchies
- Designing a professional dashboard layout
- Creating custom slicer and icon styling

These challenges helped improve understanding of:

- DAX calculations
 - Data modeling
 - SQL analysis
 - Business intelligence design principles
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Conclusion

This project successfully demonstrates how SQL and Power BI can be combined to build a complete business intelligence solution for e-commerce analytics.

The dashboard provides actionable insights related to:

- customer retention
- revenue growth
- seller performance
- category contribution
- operational efficiency

The project highlights practical skills in:

- SQL analytics
 - data modeling
 - dashboard development
 - DAX calculations
 - business storytelling
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Future Improvements

Potential future enhancements include:

- Real-time data integration
 - Predictive sales forecasting
 - Customer segmentation using RFM analysis
 - Recommendation systems
 - Deployment using Power BI Service
 - Mobile-responsive dashboard optimization
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